PROJECT ON

AZURE VIRTUAL NETWORK PEERING

CLOUD INFRASTRUCTURE | AZURE NETWORKING

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PLATFORM: MICROSOFT AZURE

REGION: EAST US 2/EAST US

PROJECT TYPE: HANDS-ON LAB

STATUS: COMPLETED

PROJECT OVERVIEW

Cloud networking is an essential component of cloud infrastructure because it enables communication between resources deployed within a cloud environment.

Microsoft Azure provides Virtual Networks (VNets) to create logically isolated network environments where resources such as virtual machines can communicate using private IP addresses.

In this project, I created two Virtual Networks in the East US 2 and East US region configured Virtual Network Peering to establish connectivity between them. I subsequently deployed two virtual machines and tested communication between them using the ping command.

This practical exercise provided hands-on experience with Azure networking, virtual machine deployment, network connectivity verification, and cloud resource management.

Project Aim:

The aim of this project was to configure communication between two separate Azure Virtual Networks using VNet Peering and verify connectivity between virtual machines deployed within the networks.

Project Objectives:

The objectives of this project were to:

- Create two Azure Virtual Networks.
- Deploy virtual machines within the respective networks.
- Configure VNet Peering between the two VNets.
- Verify the peering connection status.
- Test communication between virtual machines using pings.
- Understand private network connectivity in Azure.

TOOLS AND TECHNOLOGIES:

Tool/Technology	Purpose
Microsoft Azure	**Cloud infrastructure platform**
Resource Group	**Logical Grouping and Managment of Azure resources**
Azure Virtual Network	**Creation of isolated network environments**
VNet Peering	**Establishing connectivity between VNets**
Azure Virtual Machines	**Hosting virtual machines**
Linux	**Operating system used in the VM environment**
Azure Portal	**Creating and configuring resources**
Azure CLI	**Managing cloud resources**
Ping	**Testing network connectivity**

Creating an Azure Resource Group:

Before deploying the network infrastructure, I created an Azure Resource Group through the Azure Portal.

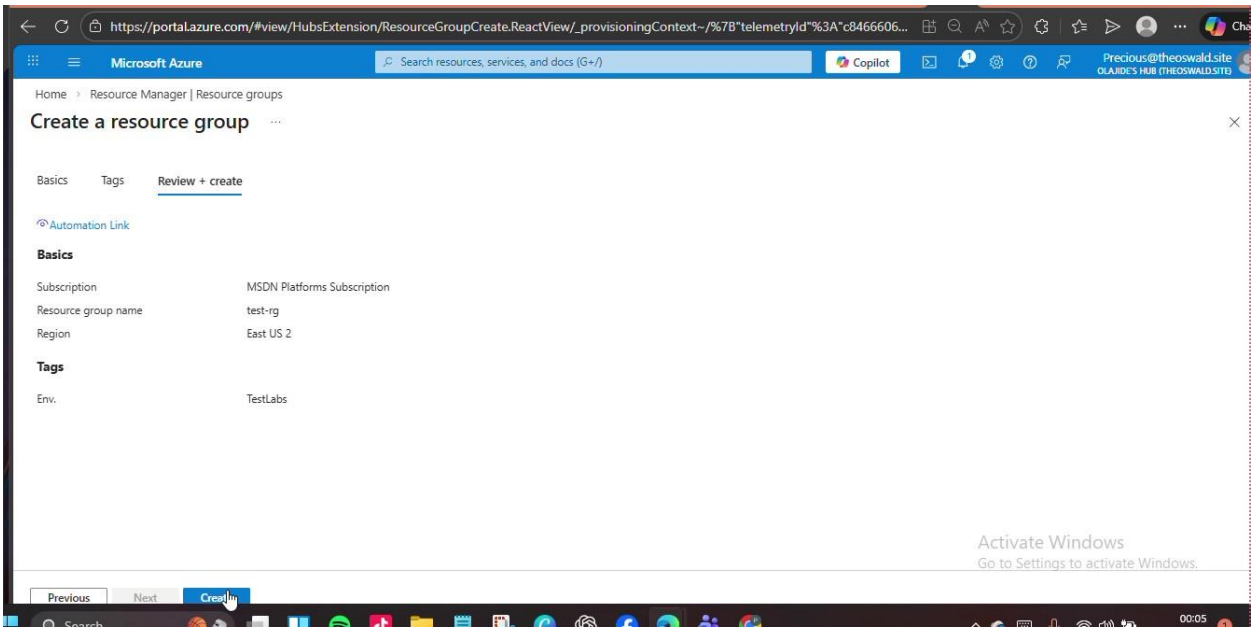
An Azure Resource Group is a logical container used to organize and manage related resources deployed within Microsoft Azure. It provides a common management boundary for resources that belong to a particular project or workload.

For this practical exercise, I created a dedicated Resource Group to keep the resources associated with the Global Virtual Network Peering project organized in one place. This included the virtual networks, virtual machines and other supporting network resources.

Using a dedicated Resource Group for a testing project makes resource management easier, improves organization, and simplifies cleanup after practical exercise. It also allows related resources to be managed as a group rather than individually.

After creating the Resource Group, I used it as an organizational container for the resources deployed during this project.

Fig.1: Resource Group created through the Microsoft Azure Portal for the Global VNet Peering project.



Network Architecture:

The lab environment consisted of two Azure Virtual Networks configured between the East US 2 and the East US region. Each virtual network hosted a virtual machine, and VNet Peering was configured to establish private connectivity between the two networks.

The architecture allowed me to examine how resources deployed in separate virtual networks communicate through Azure's virtual networking infrastructure.

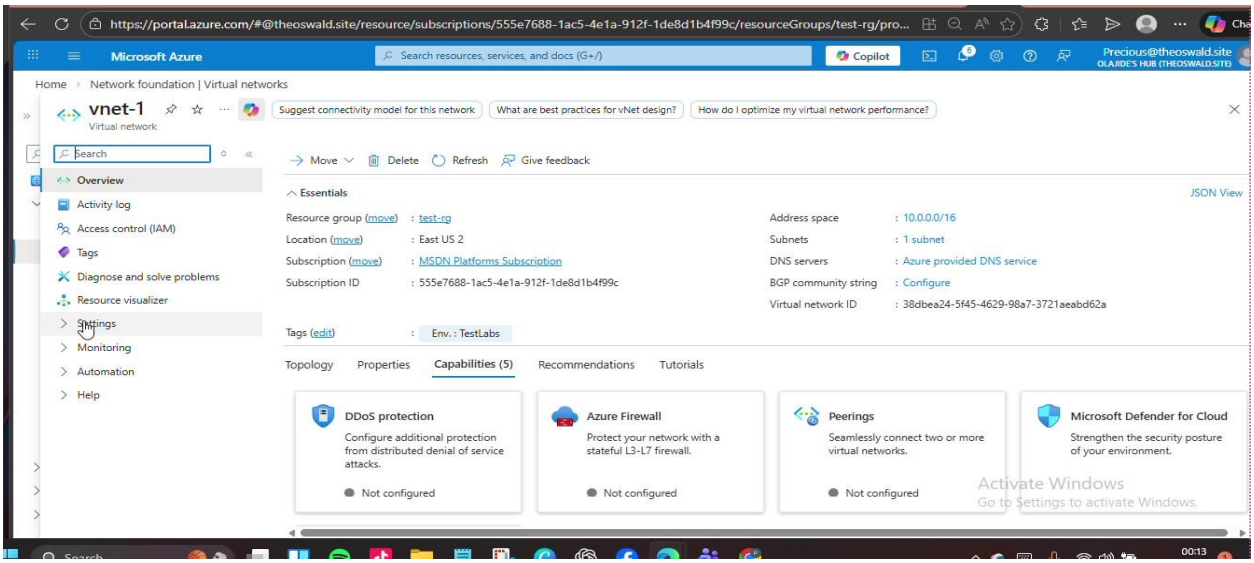
Creating the Azure Virtual Networks:

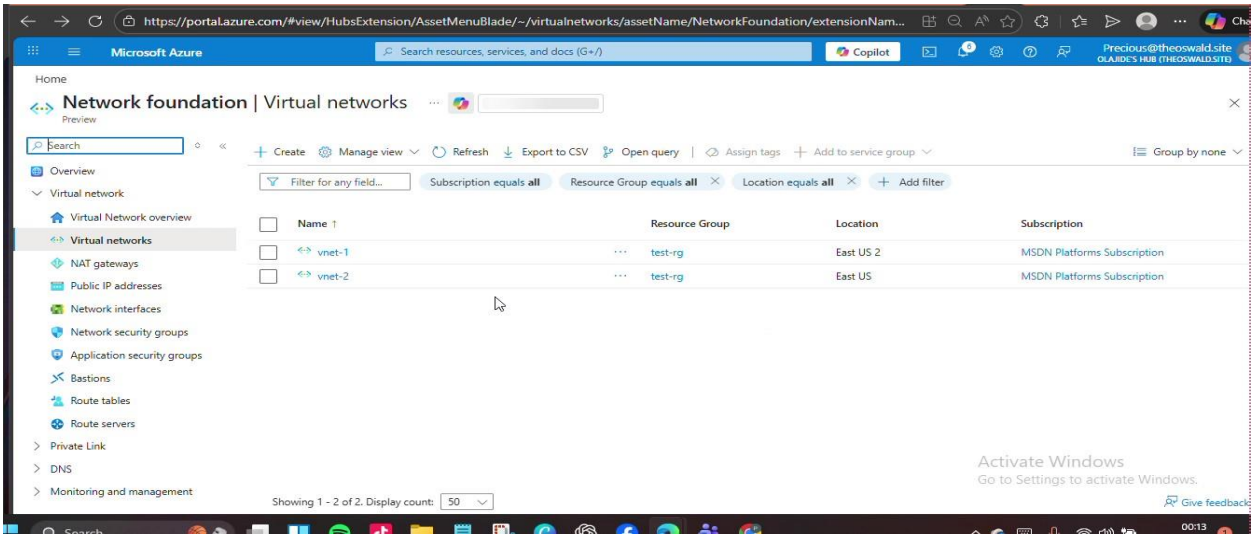
The implementation began with the creation of a Resource Group through the Azure Portal to organize the resources used in the project.

I subsequently created two separate Virtual Networks, VNet1 and VNet2, within the East US 2 region and East Us region

Each virtual network was configured with its own network address space, providing separate logical network environments for the deployment of virtual machines.

Fig.1: Azure Virtual Networks overview showing the two Vnet created in the East US region


https://portal.azure.com/#@theoswald.site/resource/subscriptions/555e7688-1ac5-4e1a-912f-1de8d1b4f99c/resourceGroups/test-rg/pro...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Precious@theoswald.site
OLAJIDE'S HUB (THEOSWALD.SITE)
Home
Network foundation | Virtual networks
vnet-1
Virtual network
Suggest connectivity model for this network
What are best practices for vNet design?
How do I optimize my virtual network performance?
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Resource group (move) : test-rg
Location (move) : East US 2
Subscription (move) : MSDN Platforms Subscription
Subscription ID : 555e7688-1ac5-4e1a-912f-1de8d1b4f99c
Address space : 10.0.0.0/16
Subnets : 1 subnet
DNS servers : Azure provided DNS service
BGP community string : Configure
Virtual network ID : 38dbea24-5f45-4629-98a7-3721aeabd62a
Tags (edit) : Env. : TestLabs
Topology
Properties
Capabilities (5)
Recommendations
Tutorials
DDoS protection
Configure additional protection from distributed denial of service attacks.
Not configured
Azure Firewall
Protect your network with a stateful L3-L7 firewall.
Not configured
Peerings
Seamlessly connect two or more virtual networks.
Not configured
Microsoft Defender for Cloud
Strengthen the security posture of your environment.
Activate Windows
Go to Settings to activate Windows.
00:13


https://portal.azure.com/#view/HubsExtension/AssetMenuBlade/~/virtualnetworks/assetName/NetworkFoundation/extensionNam...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Precious@theoswald.site
OLAJIDE'S HUB (THEOSWALD.SITE)
Home
Network foundation | Virtual networks
Preview
Search
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Resource Group
Location
Subscription
vnet-1
test-rg
East US 2
MSDN Platforms Subscription
vnet-2
test-rg
East US
MSDN Platforms Subscription
Showing 1 - 2 of 2. Display count: 50
Activate Windows
Go to Settings to activate Windows.
Give feedback
00:13

Network Configuration:

Configuration	VNet 1	VNet 2
Region	East US 2	East US
VNet name	Vnet-1	Vnet-2
Address space	10.0.0.0	10.0.1.0
Subnet	10.0.0.0/24	10.0.1.0/24

Tip: Non-Overlapping IP Address Spaces

*When creating Virtual Networks that need to communicate through VNet Peering, their IP address spaces **must not** overlap. Each VNet should have a unique address range to prevent IP addressing conflicts and ensure proper routing between the networks.*

For example, if VNet 1 uses 10.0.0.0/16 and VNet 2 also uses an overlapping range such as 10.0.0.0/24, communication and routing between the networks may fail because Azure cannot reliably distinguish the overlapping destination addresses.

Therefore, I ensured that the address spaces assigned to both Virtual Networks were distinct before configuring peering.

Azure does not allow VNet peering to be established between virtual networks with overlapping address spaces. This is why checking the address ranges before configuring peering is essential.

Configuration of Global Virtual Network Peering:

After creating the two Virtual Networks in different Azure regions, I configured **Global Virtual Network Peering** to establish private connectivity between them.

VNet 1 was deployed in the East US 2 region, while VNet 2 was deployed in the East US region. Since the virtual networks were in different Azure regions, Global VNet Peering was required to connect them.

I configured the peering relationship between the two VNets through the Azure Portal and reviewed the peering connection status to confirm that the connection was successfully established.

Fig.3a: Azure Vnet Peering Configuration showing the connection status.

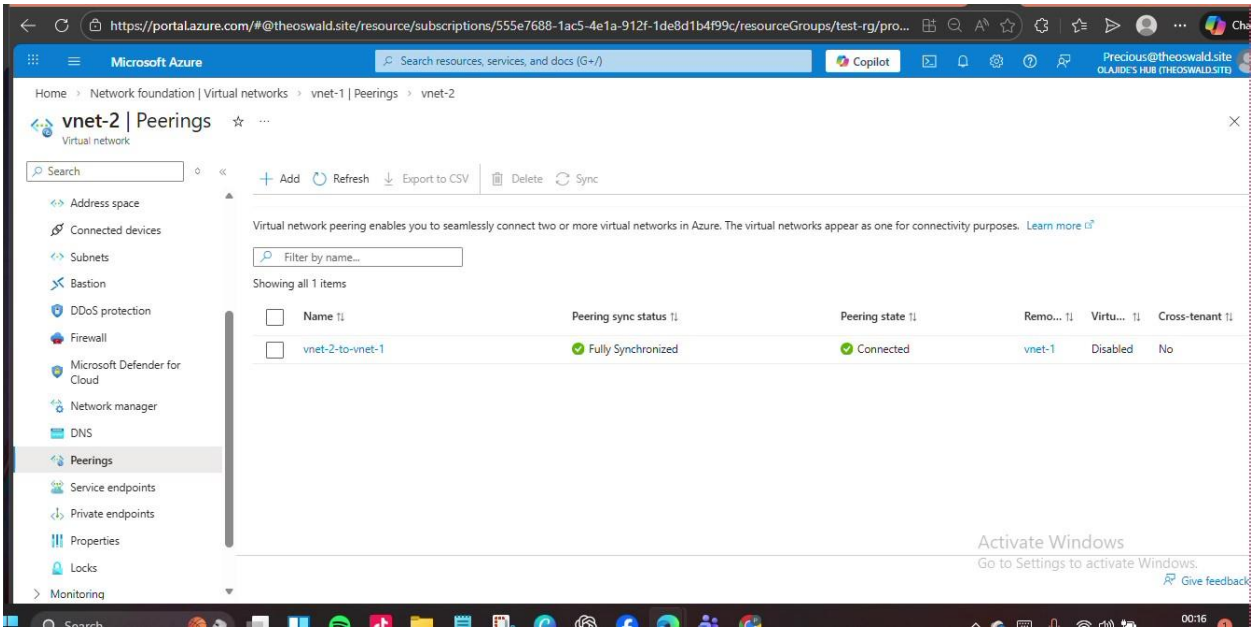
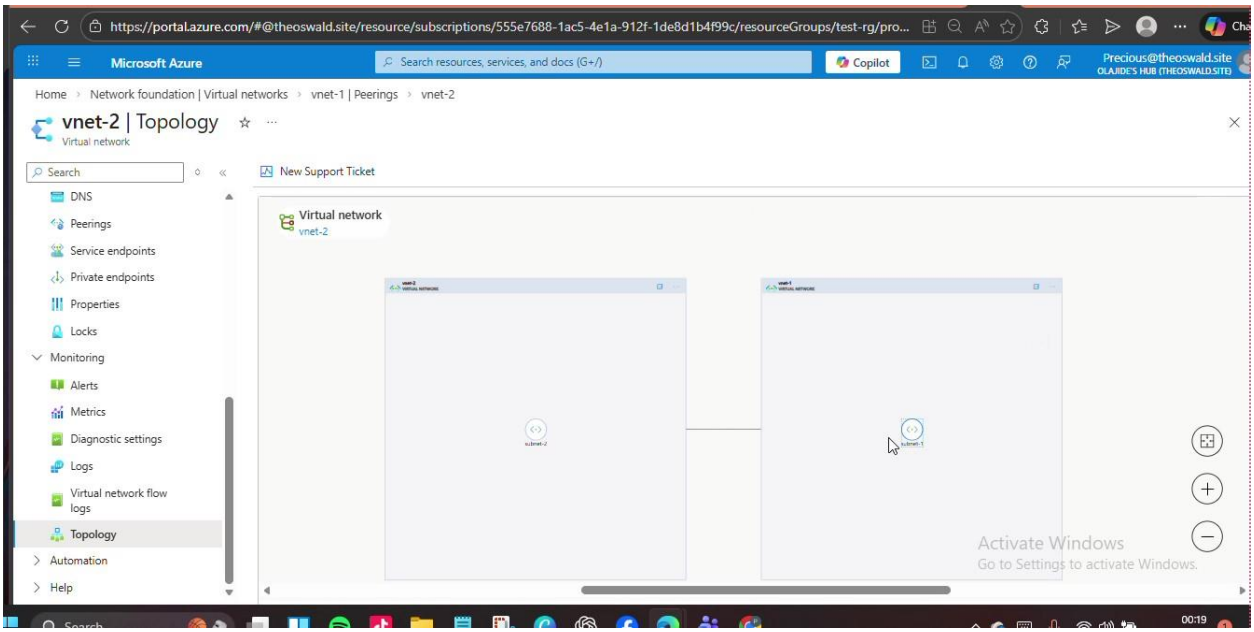


Fig.3b: Successfully established Global Vnet Peering between Vnet1 (East US 2) and Vnet2 (East US)



Peering Configuration Outcome:

The peering configuration established a private connectivity relationship between the two virtual networks, in two different Azure regions. This provided the network foundation required communication between virtual machines deployed in the respective networks.

Virtual Machine Deployment:

As part of this practical exercise, I deployed two Linux-based Virtual Machines within the Azure environment. VM1 was deployed in VNet 1, located in the East US 2 region, while VM2 was deployed in VNet 2, located in the East US region.

These virtual machines served as the compute resources for testing private network connectivity across the two Azure regions after configuring Global VNet Peering.

The deployment provided practical experience in creating Linux-based cloud resources and understanding how virtual machines communicate through peered Azure Virtual Networks.

Fig.4a: Linux Virtual Machine deployment in Vnet1 (East US 2)

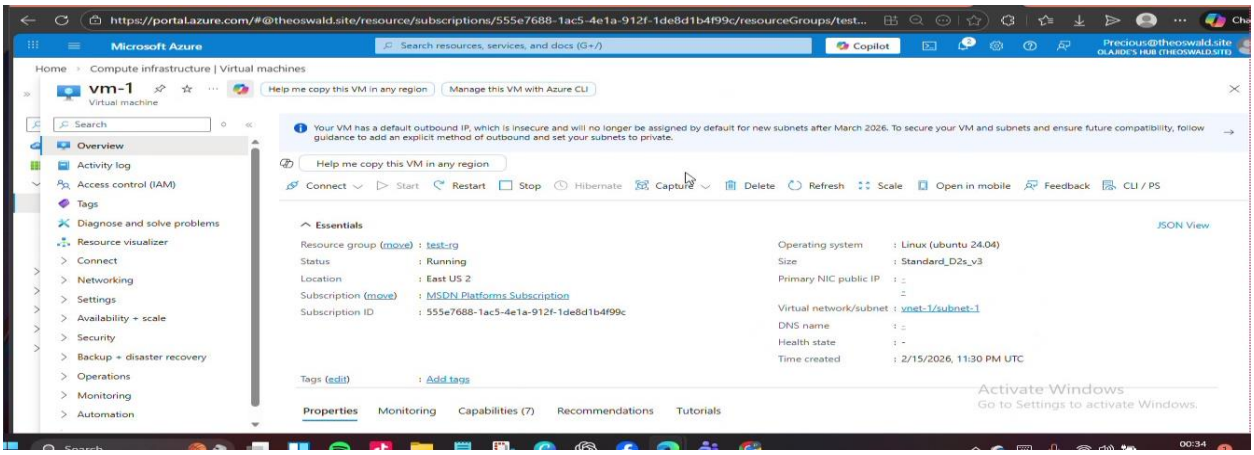
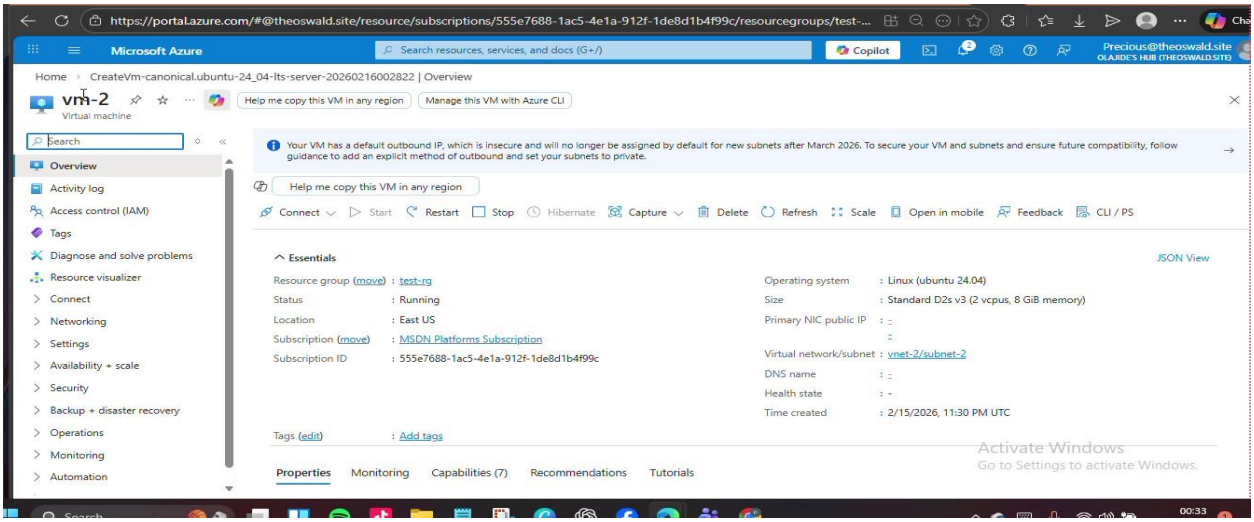


Fig.4b: Linux Virtual Machine deploymet in Vnet2 (East Us)



Virtual Machine Configuration:

Configuration	**VM1**	**VM2**
Operating system	Linux OS	Linux OS
Azure region	East US 2	East US
Virtual Network	VNet 1	VNet 2
Private IP	10.0.0.5	10.0.1.4
Connectivity	Azure Baston	Azure Baston

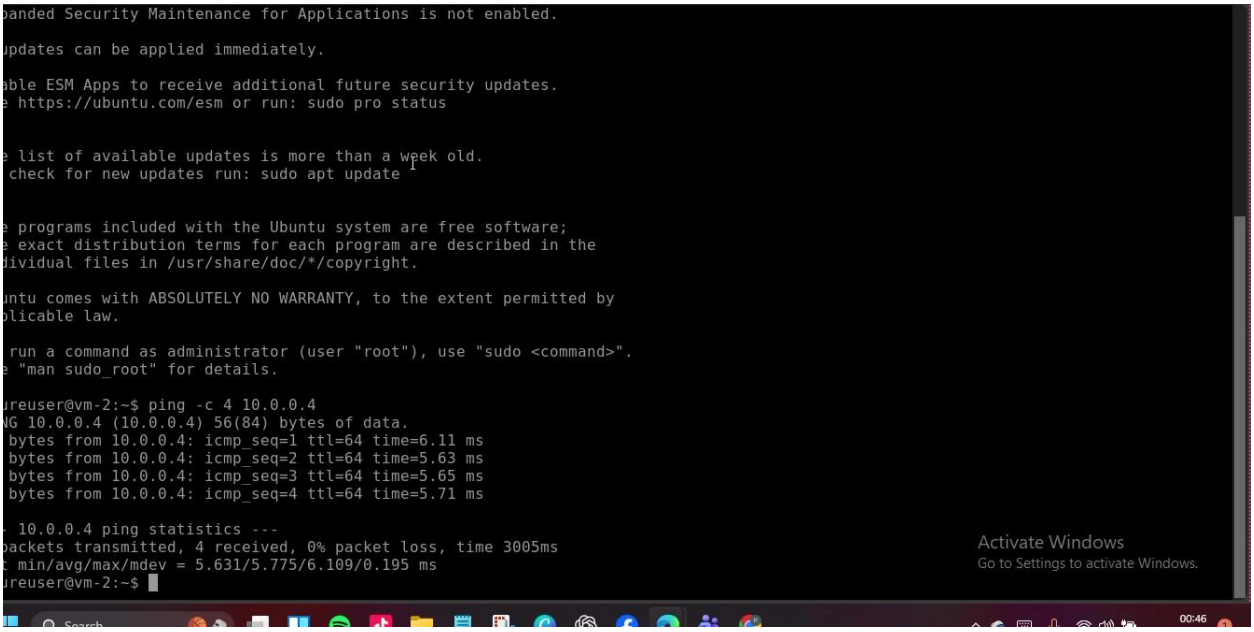
Virtual Machine Connectivity Testing:

After establishing VNet Peering, I tested network connectivity between VM1 and VM2.

I initiated a ping command from VM1 to the private IP address of VM2 to examine whether communication could take place across the peered virtual networks.

The test returned successful replies, demonstrating that the virtual machines could communicate across the configured network connection.

Fig.5: Successful ping test from VM2 to VM1 Using the private IP address



Connectivity Test Results:

The successful ping response confirmed network reachability from VM2 to VM1 at the time of testing.

This demonstrated the practical application of VNet Peering and improved my understanding of private IP communication between resources deployed in separate Azure Virtual Networks.

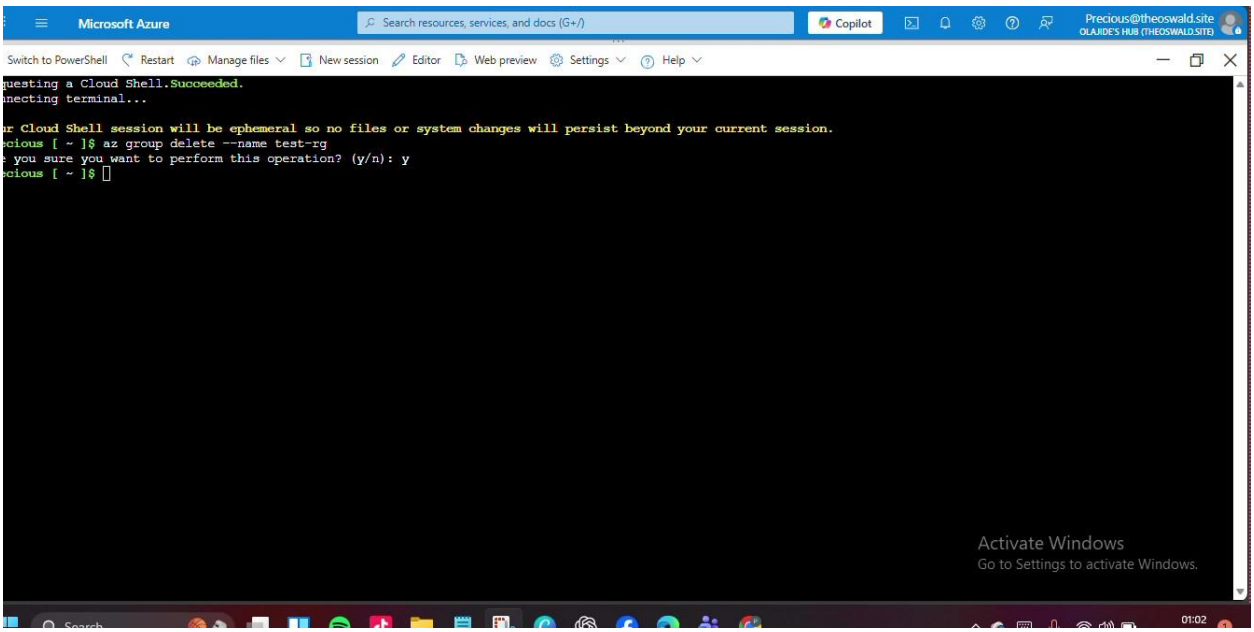
Resource Management and Cleanup:

After completing the practical exercise and capturing the necessary project evidence, I used Azure CLI to delete the Resource Group created for the lab.

Although the Resource Group was initially created through the Azure Portal, the cleanup operation was performed using the command-line interface.

This provided additional practical experience with Azure resource lifecycle management.

Fig.6: Azure CLI command used for Resource Group cleaup



PROJECT OUTCOME:

The project successfully demonstrated the configuration of two Azure Virtual Networks and the establishment of a peering relationship between them.

Connectivity testing using ping confirmed communication between VM1 and VM2 through their private IP addresses.

The exercise provided practical exposure to Azure networking, virtual machine deployment, and cloud resource management.

SECURITY CONSIDERATIONS:

The project highlighted the importance of network security when connecting cloud resources.

- Applying Network Security Groups (NSGs) to control inbound and outbound traffic.
- Limiting access to necessary ports and protocols.
- Avoiding unnecessary public exposure of virtual machines.
- Applying the principle of least privilege.
- Monitoring network activity and reviewing access configurations.

VNet Peering provides private connectivity, but it does not automatically enforce all required security restrictions.

CONCLUSION

This project strengthened my practical understanding of Microsoft Azure networking, particularly the creation of Virtual Networks and configuration of VNet Peering.

By deploying virtual machines and successfully testing connectivity between them, I was able to observe how private network communication works across peered Azure Virtual Networks.

The exercise also strengthened my confidence in using the Azure Portal and Azure CLI to deploy, inspect, and manage cloud resources.

As a Cybersecurity student building my cloud engineering skills, this project has provided a practical foundation for exploring more advanced areas of cloud networking, infrastructure security, and access management.

THE END